The DICTO Ogdoecontahedron on the Gyrated Octet

Twins at every layer: the Ogdoecontahedron, its OCD Stella and the DICTO Belted Triacontahedron in the gyrated tetrahedral–octahedral honeycomb, with one cell kit for every stacking



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Version 1 · 20 October 2026 · DICTO DOI: 10.5281/zenodo.23484012 · Companion to The DICTO Ogdoecontahedron on the Octet Truss (DICTO DOI: 10.5281/zenodo.23457016) · Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra (research branch, studies/2026-10-19-og-gyrated-octet)
The question whether the Ogdoecontahedron can live in the gyrated octet as well as in the octet, the step to its family (OCD Stella, golden envelope, DICTO Jewel), the belted solid and its name are DICTO's. Computation, checking and drafting with Claude (Anthropic).

Abstract. In a companion paper the DICTO Ogdoecontahedron (OG) sits in every octahedron of the octet truss, its eight joint hexagons on the eight faces. DICTO asked whether it can also live in the *gyrated* octet, the tetrahedral–octahedral honeycomb stacked A B A B, in which octahedron meets octahedron and tetrahedron meets tetrahedron across every layer. It can: the OGs on both sides of a layer meet on the same joint hexagon, but each is the mirror image of the other, and since the OG has no such mirror the layers are **twin boundaries**; exactly, the twin is the OG turned 60° about the joint three-fold axis, the spinel law of crystallography. Cutting each octahedron into the OG and eight chiral caps, and keeping the tetrahedra whole, gives **one exact kit for the octet, the gyrated octet and every stacking between**; at a twin layer a left and a right cap join into a twin cap and two tetrahedra into a triangular bipyramid. The facing-star middle piece of the octet fails at twin layers, and a mirror middle piece of volume $(424 - 88\sqrt{5})/4617$ takes its place. The trait runs in the family: the OCD Stella twins in the same way, meeting its twin on a whole joint triangle; the golden envelope does not fit as it is, but cut flat at two opposite three-fold corners it becomes a solid of its own, the **DICTO Belted Triacontahedron (DBT)**, which holds the OG and fills the gyrated octet's octahedra pole to pole without overlap. The DICTO Jewel, framed by cubes, does not twin. All claims are exact in $\mathbb{Q}(\sqrt{5})$ except where marked float.

1. The gyrated octet

The octet truss (Fuller [1]) is the tetrahedral–octahedral honeycomb: regular octahedra and tetrahedra, two tetrahedra per octahedron, every octahedron face against a tetrahedron. Cut perpendicular to a three-fold axis it falls into layers one octahedron thick, each layer filling the slab between two parallel planes [2, 3]. Stacking the layers so that each is the mirror image of the one below gives the **gyrated** honeycomb, the hexagonal-close-packed (A B A B) counterpart of the face-centred-cubic (A B C) octet; any sequence of the two stackings tiles space as well, periodic or not [2, 4]. In the gyrated octet every layer boundary is a mirror plane: across it an octahedron meets an octahedron face to face, and a tetrahedron meets a tetrahedron, the pair forming a triangular bipyramid.

Frame (as in [5]). The OG at the origin, its octahedron $|x| + |y| + |z| \le R$ with $\mathbf{R = 2\varphi/3}$, volume $(64 + 32\sqrt{5})/81$; tetrahedra $(16 + 8\sqrt{5})/81$; a cell of the honeycomb (one octahedron

and two tetrahedra) (32 + 16√5)/27. A twin layer is the plane x + y + z = R and its parallels. All coordinates are pairs of rationals a + b√5; corners are solved and checked exactly, volumes are exact sums of determinants.

2. The OG at a twin layer (exact)

The same hexagon. The OG's eight joint hexagons lie on the eight faces of its octahedron [5]. Reflected in the face x + y + z = R, the OG gives the OG of the octahedron on the other side, and the two meet in the same six corners: the regular joint hexagon, side² = 2/9 − 2√5/27, covering (7 − 3√5)/2 ≈ 14.59% of the face.

A true twin. The reflected OG is not a translate of the OG (as it is in the octet and in the OG's body-centred cubic packing), because the OG, a pyritohedral solid, has no mirror plane perpendicular to a three-fold axis. Exactly, **the twin is the OG turned 60° about the joint three-fold axis**, checked with the rational rotation that turns (1,0,0) to (2/3, 2/3, −1/3). This is the spinel law of cubic crystals [6], whose twin plane is parallel to an octahedron face and whose twin operation is a half-turn about the ⟨111⟩ axis: because the OG has the 120° turn about that axis, the half-turn and the 60° turn give the same twin. Round the hexagon each edge opens into a wedge between the twins: a face beside the hexagon and its mirror image make an angle whose cosine is exactly (7 − 36√5)/109, an opening of about 47.6°.

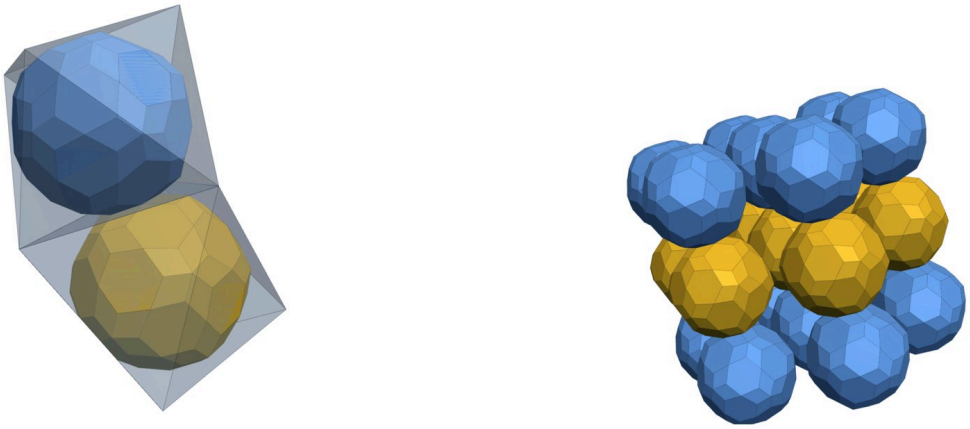


Figure 1. Left: the OG (gold) and its twin (blue) in two octahedra sharing a face; they meet on the same joint hexagon, and the twin is the OG turned 60°. Right: OGs in three layers of the gyrated octet; the layers above and below are the mirror image of the middle one.

3. One cell kit for every stacking (exact)

Cut each octahedron into the OG and **8 caps**, one under each face, by the planes through the centre and the octahedron's edges; keep each tetrahedron whole. Each cap is (octahedron − OG)/8 = **(277 − 79√5)/1539 ≈ 0.065205**, and the OG is (−1000 + 1240√5)/1539 ≈ 1.151868, (1025 − 435√5)/76 ≈ 68.83% of its octahedron. No piece crosses a face of its cell,

so **the same kit fills the octet, the gyrated octet and any stacking of the layers**, at the same density.

The caps are chiral. The OG's three-fold turn about a face's axis maps the 18 OG corners in that face's cone onto themselves, but none of the three mirrors through the axis does. So the caps come in two hands, 4 + 4 per OG: the faces (1,1,1), (1,−1,−1), (−1,1,−1), (−1,−1,1) carry one hand, the other four the other (inversion swaps them).

At a twin layer a cap meets its own mirror image, a cap of the other hand, across the part of the face outside the joint hexagon, $(3\sqrt{5} - 5)/2 \approx 85.41\%$ of the face. Fused, they make a mirror-symmetric **twin cap** of $(554 - 158\sqrt{5})/1539 \approx 0.130410$. The two tetrahedra that meet across the layer make a **triangular bipyramid** of $(32 + 16\sqrt{5})/81 \approx 0.836754$. A gyrated octet can therefore be built from the OG, the left and right caps, the twin cap and the bipyramid (and the tetrahedron, where layers stack as in the octet).

4. The facing-star middle piece at a twin layer

In the octet each octahedron edge joins two OGs whose five-fold corners face each other exactly (axes 180° apart, 0.3629 apart), and a middle piece spans them, with a thick Penrose rhombus between the stars [5, §3]. In the gyrated octet only the edges lying in a twin layer change: round them the cells run octahedron, tetrahedron | tetrahedron, octahedron, so the OG across the edge is the mirror twin, not the octet's partner. Middle pieces off the layer stay wholly on their own side (float check: no volume crosses the plane).

- **The octet's middle piece fails there** (float): about half of it (0.0399 of 0.0797) lies beyond the layer, 0.0198 of that inside the twin OG.
- **The stars no longer face** (float): the twin's five-fold corner is 0.1407 from the first, its axis turned 105.25° from it; the thick-rhombus link of the octet has no counterpart.
- **A mirror middle piece fits** (exact volume): the hull of a five-kite star and its mirror image, $(2 + 2\sqrt{5})/81$, minus the two OGs, each of which it meets in $(-155 + 101\sqrt{5})/4617$, gives **$\mathrm{Mg} = (424 - 88\sqrt{5})/4617 \approx 0.049215$**, one solid, mirror-symmetric (so with no left and right). In a float check against 43 neighbouring pieces and every OG it overlaps nothing.

So the octet's Penrose-core kit becomes a gyrated kit by adding Mg at twin-layer edges. The octahedron and tetrahedron pieces that touch a twin layer then carry different notches on that face; those boundary shapes are not computed here.

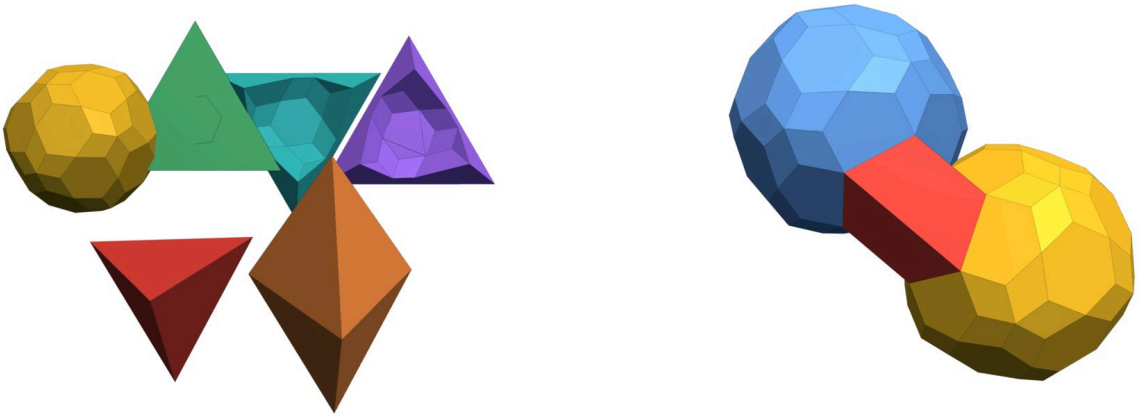


Figure 2. Left: the cell kit: the OG, the left and right caps (green, teal; the right cap shown from inside, with the OG's imprint), the twin cap (violet), the tetrahedron and the triangular bipyramid. Caps shown face-on. Right: the mirror middle piece Mg (red) between an OG (gold) and its twin (blue) at an edge of a twin layer.

5. The family (exact)

The OCD Stella (DICTO DOI: 10.5281/zenodo.23430870), the stella cut of the OG's golden companion in its body-centred cubic packing, has its eight joint faces on the same eight planes. Turned into this frame by the cube axes of that packing (the turn carries the packing's OG exactly onto the OG here), it lies in the octahedron and fills $(604 - 267\sqrt{5})/9 \approx 77.44\%$ of it. Each joint face is an equilateral triangle of side^2 $8/3 - 8\sqrt{5}/9$, $(7 - 3\sqrt{5}) \approx 29.18\%$ of the octahedron face: exactly twice the OG's joint hexagon. Its mirror twin in a face is again the Stella turned 60°, and **twin Stellas meet on the whole joint triangle**.

The golden envelope (the rhombic triacontahedron of inradius 2/3 round the OG) does not fit as it is (float): its three-fold corners poke 0.0909 beyond the faces, and across a twin layer two envelopes would overlap by about 0.0175 per shared face. In the octet those corners become keys [5, §2]; at a twin layer they have nowhere to go.

The DICTO Jewel (first released with the Kaleidohedra app, DICTO DOI: 10.5281/zenodo.23223555) does not twin (float, ray-parity sampling): in its Stella–Jewel lattice a (111) twin through a Jewel layer leaves 4.0% of the sampled points near the layer uncovered and 4.2% covered twice; through a stella layer 5.1% and 5.9%; between layers 18–19%. The Jewel and the stella octangula are framed by cubes, and a cube mirrored across a body-diagonal plane is a different cube, whereas an octahedron mirrored in its face is simply the next octahedron of the gyrated octet.

6. The DICTO Belted Triacontahedron (exact)

Cut the golden envelope flat by the two octahedron faces $x + y + z = \pm R$ on one three-fold axis. DICTO named the result the **DICTO Belted Triacontahedron (DBT)**; once cut it is a

solid of its own, not a container.

- **No new corners.** Each cut passes exactly through the pole corner's three nearest corners, so the DBT keeps 30 of the triacontahedron's 32 corners and loses only the two poles. It has 60 edges and 32 faces: 24 golden rhombi, 6 half-rhombi and 2 equilateral pole triangles of side² $8/3 - 8\sqrt{5}/9$ (side ≈ 0.824045, the same as the Stella's joint triangle).
- **Along the long diagonals.** Each cut runs along a rhombus's long diagonal (diagonal ratio² = φ^2).
- **Volume** $(-872 + 440\sqrt{5})/81 \approx 1.381110$; each removed corner $(-44 + 20\sqrt{5})/81 \approx 0.008906$, exactly **1/6 of the oblate golden rhombohedron** on the pole's three edges, $(-88 + 40\sqrt{5})/27$.
- **Symmetry $\mathbf{D_{3d}}$** (order 12): the three-fold turn, inversion and the half-turns about the three two-fold axes perpendicular to the cut axis. Sphericity 0.959 (float).
- **It holds the OG**, 36 of whose corners touch its surface.

In the gyrated octet. Every layer boundary is a twin plane, so each octahedron has its two pole faces on twin planes, exactly where the DBT is flat. Its mirror twin in a pole face is the DBT turned 60° (a true twin, as the OG). With a DBT in every octahedron, layer by translation and the neighbouring layers as mirror images, **no DBT overlaps another**: exact separating planes for all 56 neighbours checked (18 in the same layer, 19 above, 19 below). Twin DBTs meet flat, pole triangle to pole triangle, and the six keys round the belt reach into the tetrahedra of their own layer as in the octet. The DBTs fill $(493 - 219\sqrt{5})/6 \approx 55.02\%$ of space (82.53% of each octahedron).



Figure 3. Left: an OCD Stella (gold) and its twin (blue) meeting on the whole joint triangle. Right: two DICTO Belted Triacontahedra pole to pole across a twin layer, each round its OG.

7. Relation to known work, and what is claimed

The gyrated tetrahedral–octahedral honeycomb, its hcp stacking and the non-periodic stackings of the same layers are classical [2, 3, 4]; binary hard tetrahedra and octahedra self-assemble into the octet arrangement [7]. Twinning on an octahedron-face plane by a half-turn about ⟨111⟩ is the spinel law [6]; the gyrated honeycomb is described in [4] as reflective layers rotated 60°. Kabai's golden octet truss [8] is a module of one tetrahedral and one octahedral part of a golden rhombohedron, laid in planar and icosahedral lattices; his 54-faced space-filler [9] removes eight corners of a rhombic triacontahedron, each half an oblate golden rhombohedron, at the corners of a cube, for a body-centred cubic packing: a different cut and a different packing from the DBT, which removes two opposite corners, each one sixth of an oblate golden rhombohedron, and stacks as twins.

Not found in our reference library (the full text of the 80 sources in the Appendix, searched 20 October 2026) or in web searches the same day: icosahedral solids placed in the octahedra of the gyrated octet; the OG or the OCD Stella twinned across its octahedron faces; a cell kit common to the octet and the gyrated octet; the twin cap, the mirror middle piece, or the DBT as a solid. What is claimed is these results and their exact values; the honeycombs, the stackings and the twin law are not ours.

Open. The boundary pieces of the Penrose-core kit at twin layers; a fill of the DBT gyrated octet's remaining space (its tetrahedra less the six keys of each neighbour); the pieces made selectable on a gyrated-octet lattice in the apps (DICTO, planned).

8. Reproducibility

Exact scripts (ℚ(√5)): check.py (the shared hexagon, mirror ≠ translate, the wedge), twin.py (the 60° twin), kit.py and hand.py (caps, chirality), gyr_exact.py (hexagon share, twin cap, bipyramid, Mg, the DBT in the gyrated octet), stella_twin_exact.py (the OCD Stella), dbt_exact.py (the DBT). Float: gyr_cores.py (middle pieces at a twin layer), belted.py, jewel_twin.py and jewel_twin_planes.py (the Jewel). Logs: dbt-exact.log, gyr-exact.log, stella-twin.log, gyr-cores.log. Figures: gyr_page.py (meshes, float copies of the exact solids), shoot.mjs, make_figs.py.

References

1. R. B. Fuller, Synergetic building construction, US Patent 2,986,241 (1961).
2. R. Gabbrielli, Y. Jiao, S. Torquato, Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings, Phys. Rev. E 86, 041141 (2012).
3. H. Lalvani, Transformational part-count in layered octahedral-tetrahedral truss configurations, NASA TM-102635 (1990).
4. Tetrahedral-octahedral honeycomb (the gyrated form), Wikipedia, https://en.wikipedia.org/wiki/Tetrahedral-octahedral_honeycomb (accessed 20 October 2026).
5. DICTO, The DICTO Ogdoecontahedron on the Octet Truss, DICTO DOI: 10.5281/zenodo.23457016.
6. Crystal twinning (the spinel law), Wikipedia, https://en.wikipedia.org/wiki/Crystal_twinning (accessed 20 October 2026).
7. A. T. Cadotte, J. Dshemuchadse, P. F. Damasceno, R. S. Newman, S. C. Glotzer, Self-assembly of a space-tessellating structure in the binary system of hard tetrahedra and octahedra, Soft Matter 12, 7073–7078 (2016), arXiv:1605.07160.
8. S. Kabai, Golden Octet Truss, Wolfram Demonstrations Project (2007).
9. S. Kabai, 54-Faced Space-Filling Polyhedron, Wolfram Demonstrations Project (2007).

Appendix. The reference library searched

DICTO's reference library (private copies, with public citations in the Kaleidohedra repository's docs/SOURCES.md) holds 263 entries. For this paper three were added on 20 October 2026: Cadotte et al. [7] (arXiv copy) and the Wikipedia pages [4] and [6] (CC BY-SA). The full text of the 80 sources below was searched on 20 October 2026 for this paper's subjects: the gyrated honeycomb; hcp and stacking; twins and the spinel law; the tetrahedral–octahedral honeycomb and octet truss; bipyramids; the rhombic triacontahedron and its truncations; golden rhombohedra; pyritohedral symmetry; mirror images and enantiomorphs; and the names Ogdoecontahedron, OG and belted. None contains the results of this paper. The relevant hits are the honeycomb and its stackings [2, 3, 4, 7], the spinel law [6], and Kabai's golden octet truss and 54-faced space-filler [8, 9]; the word “gyrated” also occurs in a virology paper (Twarock and Luque 2019) for a two-dimensional lattice, unrelated. Thirteen further library items are page images without a text layer and could not be searched by text: the patents of Fuller (1954, 1961, 1962), Snelson (1965), Pearce (1971, 1976), Baer (1973), Hogan (1976), Bergman (1988) and Lalvani (1988, 1995, 1997), and Fedorov (1885); their subjects are recorded in the library index.

Web searches (20 October 2026): polyhedra inscribed in the octahedra of the gyrated tetrahedral–octahedral honeycomb, and twins in hcp stackings; the spinel law as a 60° turn or a (111) mirror; icosahedral space-filling solids in the octet or at its twin boundaries with the golden ratio; and the bibliographic details of [7].

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
Z. Papadopolos, O. Ogievetsky	1999	On quasiperiodic space tilings, inflation and Dehn invariants
R. V. Moody	2000	Model sets: a survey
M. Senechal	2004	Donald and the golden rhombohedra
S. Kabai	2007	54-Faced Space-Filling Polyhedron (Wolfram Demonstration; description and screenshot)
S. Kabai	2007	Golden Octet Truss (Wolfram Demonstration; description and screenshot)
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
S. Kabai	2008	A Rhombic Dodecahedron Made of Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
S. Kabai	2008	Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron (Wolfram Demonstration; description and screenshot)
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. Koskinen	2011	Unusual tilings and transformations
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
R. Gabbrielli, Y. Jiao, S. Torquato	2012	Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings
S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric

K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
S. Kabai	2015	Cluster of 14 Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
A. T. Cadotte, J. Dshemuchadse, P. F. Damasceno, R. S. Newman, S. C. Glotzer	2016	Self-assembly of a space-tessellating structure in the binary system of hard tetrahedra and octahedra
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures
N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian et al.	2020	Bi-axial woven tiles: interlocking space-filling shapes based on symmetries of bi-axial weaving patterns
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian, M. Ebert, J. Cui, C. Fu, C. Starrett	2022	Geometrically interlocking space-filling tiling based on fabric weaves
B.-X. Lao, S. Rychkov	2023	3D Ising CFT and exact diagonalization on icosahedron: the power of conformal perturbation theory
D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss	2023	An aperiodic monotile
R. Akpanya et al.	2024	Topologically interlocking blocks inside the tetroctahedrille
I. J. Dejter, L. R. Fuentes, C. A. Araujo	2025	Perfect truncated-metric codes in lattice graphs and superlattice graphs
S. Rychkov	2025	Conformal bootstrap: from Polyakov to our times
M. Kovačević	2026	An isodiametric theorem and lattice diameter-perfect codes in A3
R. Al Raisi, N. O. Koca, M. Koca, R. Koc	2026	Modified Mosseri-Sadoc tiles from D6
Wikipedia contributors	2026	Crystal twinning (incl. the spinel law)
Wikipedia contributors	2026	Fullerene
Wikipedia contributors	2026	Goldberg polyhedron
Wikipedia contributors	2026	Kenshiro Abbe
Wikipedia contributors	2026	Kyūshindō
Wikipedia contributors	2026	Mitsusuke Harada
Wikipedia contributors	2026	n-flake (Sierpinski icosahedron)
Wikipedia contributors	2026	Tetrahedral-octahedral honeycomb (incl. the gyrated form)